

Production & Distribution Rules

Student Register Suppliers — Kota City

Rule 1. Production Schedule

Production happens only on five (5) days of the week, viz. Monday to Friday. No production is carried out on Saturday or Sunday under normal operating conditions.

Rule 2. Effect of Holidays on Production

If on any production day there is a holiday, the impact on that week's production depends on whether the holiday is granted or not:

- If the holiday is NOT granted: weekly production falls by 10% for each such holiday day.
- If the holiday IS granted: weekly production falls by 20% for each such holiday day.

The reduction is applied cumulatively when more than one holiday falls in the same week (e.g., two non-granted holidays in one week reduce that week's production by 20%).

Rule 3. Effect of Holidays on Distribution

Distribution is carried out on the days specified by the company for each supplier (see the supplier list). If a holiday falls on a scheduled distribution day:

- If the holiday is NOT granted: distribution on that day falls by 50%.
- If the holiday IS granted: no distribution is done on that day.

Rule 4. Carry-Forward of Shortfall in Distribution

If on any scheduled distribution day the distribution is not done at all, or is done in a quantity less than what was required, the remaining (undistributed) quantity is carried forward.

This carried-forward quantity is added to the distribution scheduled on the first distribution day of the following week, in addition to that day's normal scheduled quantity.

Rule 5. Rotating Week Cycle

The four-week cycle (Week 1 → Week 2 → Week 3 → Week 4) rotates continuously, independent of the calendar month. Immediately after Week 4 ends, the next week is treated as Week 1, and the cycle continues in this manner indefinitely.

The cycle is anchored to Monday, 6 April 2026, which is taken as the first day of Week 1. All subsequent week labels are determined by counting forward from this anchor date in blocks of seven days, with every fifth week resetting back to Week 1.

- 06 Apr 2026 (Mon) – 12 Apr 2026 (Sun): Week 1
- 13 Apr 2026 (Mon) – 19 Apr 2026 (Sun): Week 2
- 20 Apr 2026 (Mon) – 26 Apr 2026 (Sun): Week 3
- 27 Apr 2026 (Mon) – 03 May 2026 (Sun): Week 4
- 04 May 2026 (Mon) – 10 May 2026 (Sun): Week 1 (cycle restarts)

Starting condition: As on 6 April 2026, it is assumed that there are no pending distribution shortfalls or carry-forwards from any prior week. In other words, the week immediately preceding 6 April 2026 is treated as having had no holidays, full production, and full distribution. All chained simulations and shortfall computations begin from this clean baseline.

Quick Reference Summary

Scenario	Holiday Not Granted	Holiday Granted
Production day(Mon–Fri)	Weekly production falls by 10% per affected day	Weekly production falls by 20% per affected day
Distribution day(as scheduled)	That day's distribution falls by 50%	No distribution on that day

These rules apply in conjunction with the supplier list, weekly production split, and daily distribution percentage files.